Ayush Kansal

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EDUCATION

University of Illinois Urbana-Champaign — *Bachelor of Science in Computer Science* Expected May 2029
- Relevant Coursework: Discrete Structures (CS 173), Linear Algebra (MATH 257), Calculus III (MATH 241).

University of California, Berkeley — *concurrent coursework* Jun 2024 – Aug 2025
- CS 61A Structure and Interpretation of Computer Programs (A–); CS 61BL Data Structures (A); Data C8 Introduction to Data Science (A).

TECHNICAL SKILLS

- **Languages:** Python, Java, C, Swift, JavaScript
- **CS Fundamentals:** data structures, algorithms, object-oriented design, discrete math, linear algebra
- **Systems & Tools:** Linux/Unix, Git, Bazel, FreeRTOS, CAN bus, Xcode
- **ML & AI:** PyTorch, NumPy, LLM integration (Claude API, Apple Foundation Models), computer vision, speech recognition

PROJECTS

- **Lotus Bud Academy Daycare Platform** (lotusbudacademy.com/software) — sole developer of a full-stack iOS, Android, and web platform in production: daily reports, photos, attendance, messaging, invoicing, and an AI assistant.
- **Edge ML Inference Benchmark** (github.com/AtomicAyush/edge-ml-bench; Core ML, ONNX Runtime, Swift) — benchmarked YOLOv8n at FP32/FP16/INT8 on CPU, GPU, and Neural Engine (8× faster at equal mAP); fixed INT8's 0 mAP and closed 78% of its accuracy gap via per-channel calibration and mixed precision; iPhone app: 1.6 ms image-to-boxes.
- **NoteFlash** (iOS/iPadOS; SwiftUI, SwiftData, Foundation Models) — flashcard app that turns notes, scanned pages, PDFs, slides, and Google Drive files into decks, writing cards on-device with Apple Intelligence and re-syncing changed cards from linked Drive files.
- **BTC Survivor** (Python, NumPy) — always-on service that ingests live Coinbase order-book and trade streams, turns them into features every minute, and runs 20 learning bots, with checkpoint/resume, a replay harness, a read-only dashboard, and a systemd unit; runs 24/7 on a Linux server and passed a memory soak test (paper trading only).
- **Volt** (macOS; SwiftUI, AppKit, IOKit, CoreBluetooth) — battery menu bar app reading live current and voltage from the SMC, iPhone and iPad battery over Bluetooth LE, and AirPods levels decoded from Continuity broadcasts.

ENGINEERING EXPERIENCE

Illini Electric Motorsports — *Firmware Developer, Battery Management System* Sep 2026 – Present
- Wrote and shipped real-time C firmware on FreeRTOS (built with Bazel) for the race car's battery management system: a 1 Hz CAN heartbeat, bench-verified with a CAN bus analyzer, and a charging algorithm for the 140-cell pack that tapers current as the highest cell nears its voltage limit.

Lab AK — *Co-Founder and Software Engineer* Jan 2026 – Present
- Led development of Model mAKe Smith (labak.ai): ports, runs, and compares AI models on remote edge hardware through chip-vendor toolchains (Cadence NeuroWeave), with side-by-side speed and accuracy results.
- Led development of Model Analyzer (mma.labak.ai): inspects ONNX, TFLite, Keras, PyTorch, and TensorFlow models layer by layer, with weights and multiply-accumulate cost, entirely in the browser.

PilotCity — *Software Developer; Data Scientist* Nov 2022 – Aug 2024
- Fixed defects before deployment, optimized code paths, and cleaned and analyzed large datasets.

Seeed Studio — *Intern* Jun – Aug 2023
- Built edge-AI software for microprocessors, including an AI subject-tracking camera that kept tracking off-frame.

Alan AI — *Intern* Jul – Sep 2022
- Built a voice-activated GitHub FAQ on Alan AI's speech recognition platform, targeting a ~30% reduction in support call wait times.

RESEARCH EXPERIENCE

ASDRP — *Lead Researcher, Author, and Developer* Jun 2024 – Jan 2026
- Co-built ASDRP's iOS app (SwiftUI, Firebase; owned the student profile system) and built a Google Chat reminder bot that checks status-update quality with GPT.
- Co-first author of a paper under review at the *Journal of Emerging Investigators*; built the Python pipeline that processed 150 hours of audio across four speech recognition systems; at 1.25× speed (20% cheaper), Azure and Deepgram stayed under 0.10 WER.

Osteoporosis Gait Sensor — *Lead Developer and Designer* Jun 2024 – Present
- Designed and prototyped a wearable foot sensor that uses machine learning to detect early osteoporosis indicators from gait data.
- Built the instrument around a Raspberry Pi and a 16-sensor array laid out in the shape of a foot, with a real-time collection, preprocessing, and model-based analysis pipeline.

LEADERSHIP

Byte By Byte Coding — *CEO and Co-Founder* Dec 2024 – Present
- Founded a 501(c)(3) teaching computer science to 50+ underserved students; built the curriculum and lead volunteer instructors.

Boy Scouts of America — *Eagle Scout; Senior Patrol Leader; NYLT Staff* Aug 2018 – Jun 2026
- Taught leadership at National Youth Leadership Training; Presidential Volunteer Service Award (Gold) twice for 100+ service hours each.